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Reliable simulation analysis for high-temperature inrush water hazard based on the digital twin model of tunnel geological environment

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ABSTRACT

In complex mountainous terrains, tunnel construction faces unique challenges from high-temperature water inrush hazards, a systemic risk arising from the interplay of stress, seepage, and temperature fields. Traditional simulation methods, focusing on isolated disaster scenarios, fall short in addressing the multifaceted nature of these risks due to geological ambiguity and data incompleteness. Digital twin technology presents an effective solution to these challenges; however, its core challenge lies in how to utilize digital twin technology for data-model-co-driven simulation analysis of coupled multi-physical fields in situations of incomplete data. This paper introduces a digital twin paradigm for the simulation analysis of water inrush, which significantly enhances efficiency and accuracy through the integration of advanced machine learning and finite element analysis techniques. Specifically, this is achieved by combining a high-precision geological modeling method based on Gaussian Processes (GP) with a parameter calibration method through Gaussian Process-Differential Evolution (GP-DE) back-analysis. Firstly, a voxel structure is utilized to integrate the multi-field attribute features of the tunnel environment. Secondly, through the integration of multi-source advanced geological prediction data, we construct a dynamic digital twin model of the tunnel environment leveraging machine learning techniques. To overcome the issue of low modeling accuracy, the GP is employed, enhancing the exploitation of latent information within multi-source geophysical data. Lastly, we utilize the GP-DE back-analysis method to calibrate the parameters of the tunnel environment, thereby enhancing the precision and reliability of water inrush simulations. The method has been validated through application to a section of an ultra-high-temperature water inrush tunnel in China, featuring a burial depth of 230 meters. The accuracy of the method is corroborated by the monitoring data from the tunnel, supporting dynamic optimization design and safety prevention measures during construction.

1. Introduction

In the advancement of railway construction in mountainous regions, tunnels constitute more than 80% of the total infrastructure, with many being deeply buried and facing disaster risks such as high geostress, high geothermal conditions, and sudden water inrush (Li et al., 2021a). Typically, engineers depend on geophysical and sensor data to predict the geological conditions ahead of the tunnel face and to formulate the corresponding construction plans. Owing to the concealed and spatiotemporally heterogeneous nature of the tunnel environment, this setting transforms into a complex and open spatial environment (Wang et al., 2021). In actual construction, engineers often depend on the interpretation of advanced geological prediction data to forecast the geological conditions, attributable to the limitations of detection methods (Cui et al., 2015; Huang et al., 2022; Xin-rong et al., 2015). The four predominantly utilized advanced geological detection technologies include: Tunnel Seismic Prediction (TSP), Transient Electromagnetic Method (TEM), Induced Polarization (IP), and Ground Penetrating Radar (GPR). The differences primarily reside in data formats, dimensions, and detection ranges. The heterogeneity of data frequently results in divergent interpretation outcomes when utilizing a single data source (Li et al., 2014). Furthermore, the coarse, imprecise qualitative assessments obtained due to the inadequate extraction of potential information from the data fail to satisfy the sophisticated requirements of geological modeling. Consequently, determining how to effectively integrate multi-source geophysical data and enhance the accuracy and reliability of comprehensive interpretation and modeling of various adverse geological attributes in the tunnel environment represents the first core challenge addressed in this study.

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Additionally, during construction, risks associated with high-temperature water inrush in these geothermally active environments are more complex than traditional hazards such as high rock temperature or sole water inrush (Wang et al., 2024a). This type of thermal damage constitutes a systemic and comprehensive risk, arising from the interplay of stress, seepage, and temperature fields, which poses significant challenges in risk prevention and control during construction (Liu et al., 2023c). Existing simulation methods that focus on single disaster types are inadequate in accounting for the coupled effects between multiple disaster risks, thereby hindering accurate risk assessment. In disaster simulations, certain key parameters cannot be measured accurately, leading to a significant discrepancy between the calculated results and the actual situation. Consequently, addressing the second core challenge in this study involves constructing a thermo-hydro-mechanical (THM) coupling water inrush simulation model featuring accurate key parameters.

To address these challenges, the solution strategy must be intimately aligned with practical engineering considerations, necessitating a focus not only on precision but also on efficiency—the hallmark of digital twin (DT). Through the integration of multi-source real-time data, mechanistic models, and expert knowledge, DT provides an accurate representation of the physical world and facilitates risk assessment, prediction, and decision-making (El Saddik et al., 2021). Recently, bolstered by the integration of the Internet of Things, big data analysis, and advanced simulation algorithms, DT has seen a progressively wider application in tunnel engineering (Hu et al., 2021). Indeed, DT offers several advantages in tunnel engineering, including:

(1) DT enables the integration of multi-source observational data and facilitates the real-time updating of geological models (Wu et al., 2022). This rapid updating mechanism enhances the timeliness and accuracy of the model, enabling it to more accurately reflect the actual underground situation.

(2) DT employs advanced machine learning methods to delineate the characteristics and trends of incomplete data in the tunnel and to unveil its underlying principles (Latif et al., 2023; Erharter et al., 2022).

(3) Through the integration of models from various disciplines (Tao et al., 2022), DT is capable of accurately simulating the disaster process due to the coupling effect of multiple factors in tunnels.

Although DT has found applications in domains like tunnel rock design (Xiaojun et al., 2023), operations and maintenance (Yu et al., 2021), and lighting environment optimization (Shen et al., 2022), its deployment in tunnel geological modeling and disaster simulation remains limited. This limitation emerges due to the specialization of these methods for specific scenarios, resulting in a lack of universality. Therefore, in addressing the aforementioned core challenges, this paper proposes a geological modeling and water inrush simulation analysis method tailored for DT applications. Specifically, this study advocates for the adoption of voxel models as the foundational framework for DT. Voxel models are considered ideal owing to their ability to store multiple attributes of the geological environment and to enable efficient dynamic updates (Wu et al., 2022). The contributions of this study are as follows:

(1) A voxel model was utilized to integrate multi-source geophysical data, effectively characterizing the multi-dimensional attributes of the tunnel environment. The Gaussian Process (GP) method was employed to establish the nonlinear relationship between these attributes and adverse geological conditions. This approach addresses the limitation of traditional single data sources, which fail to fully capture the complexity of the surrounding rock, thereby enhancing the accuracy and reliability of the DT model of the tunnel geological environment.

(2) We propose a parameter calibration method based on the Gaussian Process-Differential Evolution (GP-DE) back-analysis to improve the reliability of the water inrush simulation by accurately calibrating the tunnel environmental parameters. This method addresses the high parameter uncertainty and insufficient simulation precision in traditional models when dealing with complex underground environments.

(3) We applied the proposed method to a section of a tunnel with a burial depth of 230 meters in southwest China, which is prone to ultra-high-temperature water inrush. The experimental results prove the effectiveness and superiority of our method.

The structure of this paper is as follows: Section 2 briefly reviews previous studies and summarizes the shortcomings of DT modeling of geological environments and high-temperature water inrush disaster simulation. Section 3 introduces a DT paradigm for geologic modeling and inrush water simulation analysis. Section 4 introduces the mathematical principles of geological environment DT modeling based on GP and high-temperature water inrush simulation parameter calibration based on GP-DE back-analysis. We verified the effectiveness of this method using data from a tunnel section with a burial depth of 230 meters in southwest China and discussed the results and performance in Section 5. Finally, in Section 6, we summarize the paper.

2. Related works

2.1. DT modeling of geological environments

The crux of tunnel geological DT modeling resides in constructing high-precision 3D geological models. Existing methods are broadly categorized into three types: geological statistical methods, implicit modeling methods, and geophysical exploration interpretation methods:

(1) Geological statistical methods: For modeling rock mass fractures, after collecting and statistically analyzing fracture orientation data, stochastic Discrete Fracture Network (DFN) geological models can be generated using methods such as Monte Carlo simulations. These models aim to statistically replicate the actual distribution of rock fractures (Gottron and Henk, 2021; Li et al., 2021b; Fan et al., 2023b). Additionally, borehole data surrounding the tunnel can be used in conjunction with Pluri-Gaussian Simulation (PGSIM) and Sequential Gaussian Simulation (SGSIM) to generate 3D spatial variability and uncertainty models of geological-geotechnical parameters (Gangrade et al., 2022a,b). Although geological statistical methods demonstrate significant potential, their effectiveness remains highly dependent on the integrity and representativeness of the underlying datasets (Ogbu et al., 2024).

(2) Implicit modeling methods: These methods create three-dimensional geological models by solving implicit functions to reconstruct surfaces that represent the isosurfaces of geological bodies' geometric shapes (Zhong et al., 2021). Typically, functions such as Radial Basis Functions (RBF) and Hermite functions are employed to formulate interface functions, using geological points, boreholes, and other data as constraints to determine optimal geological interfaces. Boolean operations are then used to segment different geological bodies. This approach offers several advantages, including a high degree of automation, smooth model surfaces, and the elimination of topological errors (Sun et al., 2024). However, implicit modeling faces challenges when applied to tunnel geological modeling, particularly in mountainous tunnel construction. The harsh environmental conditions and technical limitations often result in the collection of limited borehole or cross-sectional data, leading to incomplete and sparse datasets (Xiong et al., 2018). This issue is especially pronounced with small datasets, where the spatial resolution of the generated geological model may not meet the high-precision requirements essential for DT applications.

(3) Geophysical exploration interpretation methods: These methods construct geological models by observing and interpreting various changes in geophysical fields (Rezaie et al., 2017; Olierook et al., 2021; Panzera et al., 2022). In the tunneling field, techniques such as TSP and Tunnel Reflection Tomography are commonly used to determine the spatial distribution of faults, fractured zones, and underground cavities (Fan et al., 2021; Pan et al., 2020). Additionally, methods like TEM and IP are frequently employed to detect water content and delineate the spatial distribution of water-rich zones (Nie et al., 2024;

Khan et al., 2020). Each of these methods requires specialized data processing software. However, these geophysical techniques operate independently, with each method capable of interpreting and analyzing only a specific type of data. This lack of integrated interpretation leads to limited accuracy in the inversion process (Li et al., 2014).

In response to these challenges, and considering the increasing application and notable successes of machine learning in underground engineering in recent years (Huang et al., 2021), this paper presents a multi-source data fusion-based machine learning approach for tunnel geological environment modeling. By fully integrating multi-source data and uncovering hidden features, this method significantly enhances the accuracy and reliability of the modeling process.

2.2. Simulation of high-temperature water inrush

Presently, theories such as peridynamics (Gao et al., 2021), finite element method (Wang et al., 2023), discrete element method (Zhou et al., 2020), and smoothed particle hydrodynamics (Fan et al., 2023a) are utilized to simulate and analyze the spatiotemporal dynamics of water inrush disasters. To address the coupling effects of high geotemperature, osmotic pressure, and geostress, many researchers have developed various THM coupling mechanism models, including saturated porous medium THM coupling models (Zhao et al., 2020), THM coupling models that consider rock damage (Shi et al., 2023), and thermal-nonlinear hydraulic-mechanical models (Liu et al., 2018). However, the focus of most research on rock stability analysis or on the influential factors frequently does not directly guide tunnel construction (Peng et al., 2022), primarily attributed to the poor construction environment and technical limitations that result in incomplete data collection, hindering accurate determination of the boundary conditions and key parameters. Nevertheless, with the aid of the back-analysis technique, it remains possible to calibrate the numerical simulation parameters utilizing some monitoring data in the tunnel. Back-analysis involves inferring unknown environmental parameters through iterative optimization utilizing displacement, stress, and water inrush data recorded during the construction process (Berti et al., 2017; Feng et al., 2000; Kaiser et al., 1990). Optimization methods can generally be divided into deterministic and stochastic methods. Deterministic methods, including the Levenberg–Marquardt and Nelder–Mead simplex methods, often fall into local optima and are deemed unsuitable for problems with multiple inversion parameters (Li et al., 2023). Stochastic optimization methods, such as particle swarm optimization (PSO) (Chen, 2021), differential evolution (DE) (Chang et al., 2023), simulated annealing (SA) (Gutierrez, 2020), and genetic algorithms (GA) (Sun et al., 2021), have demonstrated flexibility in addressing complex nonlinear optimization challenges. Moreover, existing back-analysis studies in underground engineering typically rely on a single displacement monitoring value to calibrate mechanical parameters (Zhang et al., 2020; Luo et al., 2018). However, this approach is not suitable for mountainous tunnel environments, which are influenced by the coupling effects of high temperature, high water pressure, and high geostress. This paper proposes a GP-GE back-analysis parameter calibration method based on a THM coupled model.

3. DT paradigm for modeling and simulation analysis

This section introduces a DT paradigm for geological modeling and water inrush simulation analysis, as illustrated in Fig. 1. The concept of DT has been extensively explored and defined by numerous scholars. Building on the definition provided by Xiaojun et al. (2023), this paper adapts the DT framework to suit the specific requirements and applications of tunnel construction. The framework is divided into four key components: Digital Twin Scene (DTS), Digital Twin Entity (DTE), Digital Twin Disturbance (DTD), and Digital Twin Processor (DTP). These components are not merely theoretical constructs but are also practically applied in tunnel construction, demonstrating their real-world utility. By using tunnel construction as an illustrative example, this paper seeks to bridge the gap between conceptual DT frameworks and their implementation in engineering practice, thereby advancing the understanding and application of DT in complex geological environments.

DTS refers specifically to the application of DT technologies in the context of tunnel water inrush scenarios. It is a focused, task-specific application within a defined time and space, aimed at addressing particular engineering challenges. While a single DT environment can encompass multiple DT scenes, our DTS is dedicated to the real-time acquisition and analysis of data related to high-temperature water inrush hazards in tunnels.

The DTE component for tunnel modeling includes the tunnel structure, the geological environment (comprising rock mass fracture zones and water-rich areas), and the distribution of multiple physical fields, such as temperature and seepage fields. The construction of the tunnel structure entity is facilitated by MicroStation software, which processes the tunnel design's Building Information Modeling (BIM). The fractured zones and water-rich areas within the geological environment are modeled using machine learning techniques based on a voxel model. The distribution of multiple physical fields is depicted through voxel models obtained from numerical simulations.

Excavation represents the primary role of DTD in tunnels. Following excavation, changes in the geological environment trigger PE equipment to commence data collection. This data is then fed into the DTP for processing, geological modeling, and simulation analysis, guiding the safe construction of the tunnel.

DTP primarily handles data processing and model invocation. It involves selecting and applying appropriate models to integrate, process, and analyze input data, thereby generating computational results. In this study, after integrating data based on the voxel model, various machine learning algorithms were evaluated, and the best-performing model (as detailed in Section 4.2) was employed for geological modeling. Subsequently, the parameter calibration model (as described in Section 4.3) was applied to enhance the reliability of numerical simulations.

4. Methodology

4.1. Overview

This section delineates the reliable simulation method for high-temperature water inrush utilizing the DT model of the tunnel geological environment with comprehensive detail. Fig. 2 illustrates the workflow, which comprises two key steps:

(1) DT modeling of the tunnel geological environment: Following the integration of data within the voxel model, several machine learning algorithms were evaluated, and the best-performing model was selected for geological modeling.

(2)Reliable Simulation Analysis Utilizing the DT Model: A back-analysis method employing GP-DE is proposed, which takes into account the THM coupling to calibrate the tunnel environmental parameters. Subsequently, the multi-physical field distribution around the tunnel is calculated by simulation based on the optimal parameters, which further complements and improves the DT model.

4.2. DT modeling

i. Voxelization and integration of multi-source data.

Due to simpler topology, higher computational efficiency, and faster update speed (Wu et al., 2022; Jørgensen et al., 2013; Liu et al., 2023a), this study opts for square-shaped voxels over triangular-shaped voxels (such as triangular prism voxels). These characteristics make them particularly well-suited for high-precision geological modeling in small areas and for representing digital twin models (Aleksandrov et al.,

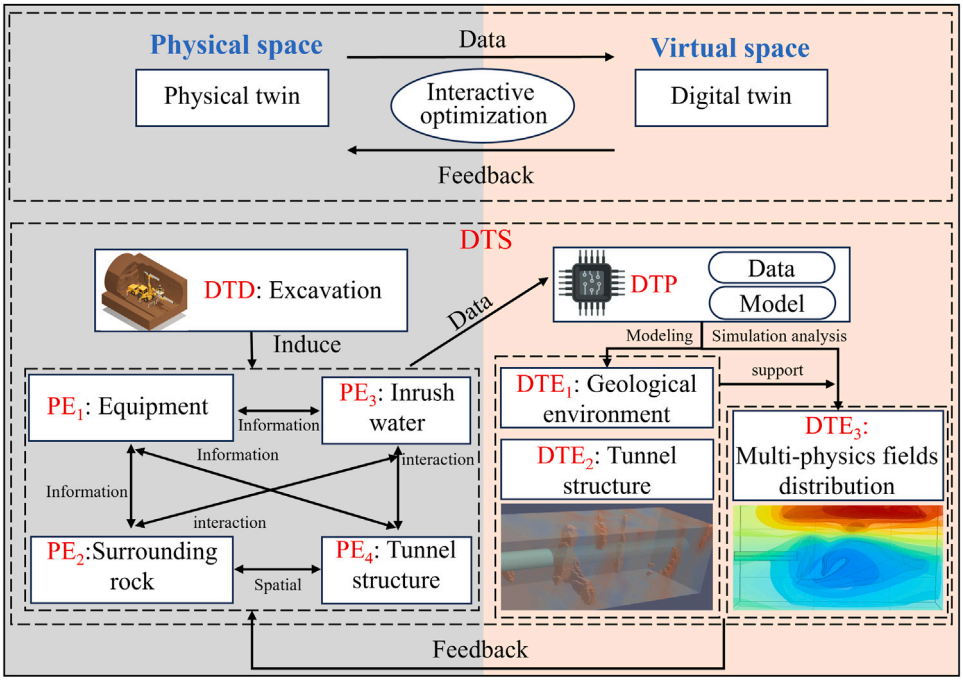


Fig. 1. DT paradigm for modeling and simulation analysis.

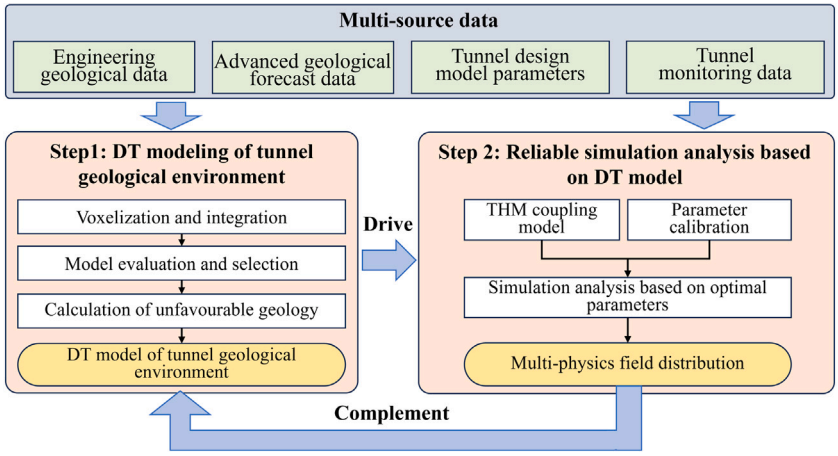


Fig. 2. Simulation and analysis workflow for high-temperature water inrush based on the DT model of tunnel geological environment.

2021). Furthermore, the original geophysical data used in this study (such as TSP) is already in a 3D regular grid format. Converting this data into triangular prism voxels would not only introduce additional complexity but could also introduce noise through interpolation or resampling, potentially compromising the accuracy and stability of the predictive model (Li and Heap, 2014).

For TSP data, TSP 303 Plus equipment along with its accompanying software is employed for data processing, producing spatial coordinates and characteristic values of geophysical parameters including longitudinal wave velocity, shear wave velocity, and Poisson's ratio. These parameters are recorded in a textual format, with the voxel model being constructed by directly assigning the values to the corresponding coordinates. For TEM data acquisition and processing, the YCS-2000 A mining transient electromagnetic instrument and its accompanying software were utilized. The exported data provided resistivity distributions across multiple planes within the spatial domain; however, these distributions were unevenly spaced, with some regions exhibiting sparse data points. The RBF interpolation method was selected for its robustness and stability under such conditions. This method is widely recognized in geological modeling, with numerous studies attesting to its reliability in producing accurate and dependable results in similar contexts (Hillier et al., 2014; Wang et al., 2018; Rusu and Rusu, 2006). By applying RBF interpolation, we achieved a smoothly continuous resistivity distribution, which was subsequently used to construct the voxel model.

The subsequent step involves the unified integration of the voxel models, which includes the following procedures: (1) Unifying the coordinate system: Extracting grid position data from the voxel models and mapping it onto a common spatial coordinate system for seamless integration. (2) Defining the modeling area: Focusing on the area around the tunnel face, the voxel models are integrated within a specified spatial range. (3) Unifying the grid resolution: Resampling the voxel models within the defined spatial range using the linear interpolation method to ensure a consistent grid resolution.

ii. Model evaluation and selection.

For quantifying rock integrity and water inrush intensity in tunnels, this study references the China national standard for the engineering classification of rock mass (Wu et al., 2014), where rock integrity is

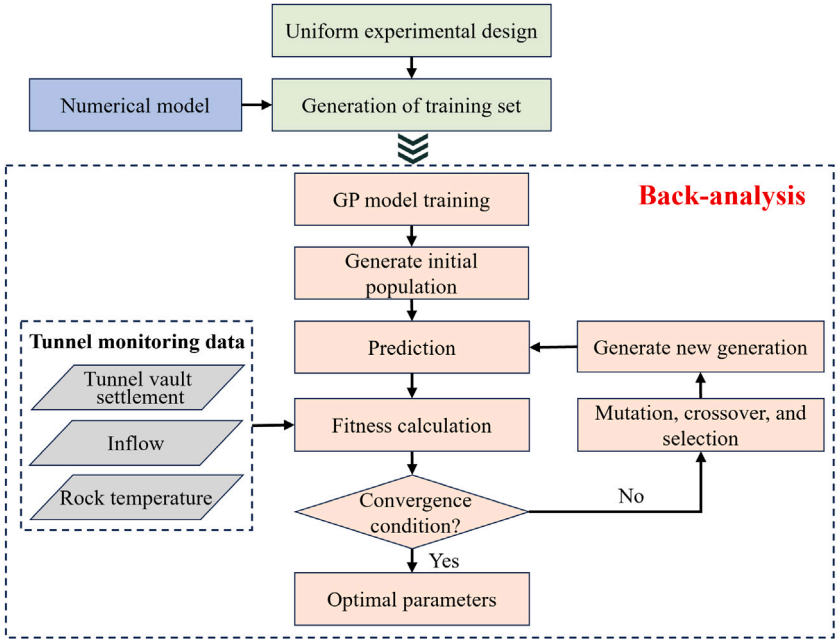


Fig. 3. Flow chart of parameter calibration.

classified into five levels: R1 (intact), R2 (partly intact), R3 (partly broken), R4 (broken), R5 (extremely broken). Water inrush intensity is delineated into four levels: W1 (dry), W2 (wet), W3 (linear water inrush), and W4 (gush water inrush). During tunnel excavation, experts manually identify the state of the rock mass at various tunnel faces, classifying rock integrity and water inrush intensity based on their judgment. To construct the sample database, corresponding physical parameters such as shear wave velocity, longitudinal wave velocity, density, Poisson's ratio, Young's modulus, and resistivity are extracted based on the voxel model.

We employed six machine learning algorithms: XGBoost, AdaBoost, CatBoost, Support Vector Machine (SVM), Random Forest (RF), and GP. These algorithms were chosen for their specific strengths in addressing the challenges associated with tunnel geological modeling. XGBoost and AdaBoost excel in boosting techniques, enhancing model accuracy by combining weak learners (Chen and Guestrin, 2016; Zhu et al., 2009). CatBoost was selected for its proficiency in handling categorical data, which is prevalent in geological datasets (Prokhorenkova et al., 2018). SVM was chosen for its robustness in managing high-dimensional data (Jian et al., 2012), while RF was included for its ability to mitigate overfitting and effectively handle a large number of input variables (Dong et al., 2013). GP was utilized for its superior performance in small sample sizes and nonlinear modeling scenarios (Zheng et al., 2022).

Each algorithm was trained on the integrated dataset, with hyperparameters optimized for optimal performance. We evaluated the models using Mean Average Precision (mAP) as the assessment metric. The model with the highest mAP score was selected for the final geological modeling. This best-model selection method ensured the adoption of the most accurate and reliable model, thereby guaranteeing the precision and broad applicability of the DT framework.

iii. Calculation of unfavorable geology.

Considering the smaller data coverage of TEM compared to TSP, and the presence of spatial areas exclusively containing TSP data, the constructed models are categorized into two groups: one utilizing only TSP data, and the other incorporating both TSP and TEM data. By doing calculations on each voxel, the method calculates the three-dimensional distribution of rock mass integrity and water inrush intensity levels within the specified area. This procedure ultimately results in the creation of a DT model representing the tunnel's geological environment.

4.3. Reliable simulation analysis based on DT model

i. THM coupling model and parameter calibration.

For the reliable simulation of high-temperature water inrush, selecting a suitably complex mechanism model is crucial because of the sophisticated interplay between the seepage field, stress field, and temperature field. This interplay involves seepage affecting both temperature and stress distribution, stress variations influencing rock permeability, and temperature changes impacting the hydrodynamic properties and mechanical behavior of the rock (Liu et al., 2023b). Therefore, a model that can accurately represent these physical interactions is essential for simulating the high-temperature water inrush process effectively. In this research, COMSOL Multiphysics is employed to develop a coupled mechanical-hydraulic-thermal model incorporating Darcy's Law for fluid flow, porous media heat transfer, and solid mechanics for structural analysis. To simulate real-field conditions while simplifying the mathematical model, the following assumptions are made: (1) The rock is considered an isotropic porous medium. (2) The fluid, exclusively water, follows Darcy's law for flow and Fourier's law for heat transfer, assuming local thermal equilibrium.(3) A uniform load is applied at the top of the model.

In addition to selecting a comprehensive mechanism model, the accuracy of simulation parameters is crucial for ensuring the reliability of results. This study utilizes a back-analysis technique, a well-established method for parameter calibration, which enables the refinement of model parameters based on observed data. Specifically, a joint back-analysis method using the GP-DE algorithm is proposed. This technique calibrates critical simulation parameters, such as elastic modulus, permeability, and geothermal gradient, by conducting a detailed analysis of key indicators, including tunnel vault settlement, water inflow, and rock temperature, as illustrated in Fig. 3.

Suppose the tunnel environmental parameters to be estimated are represented as:

$$\begin{cases} X = [x_1, x_2, \ldots, x_m]^T \\ x_{i\ \min} \leqslant x_i \leqslant x_{i\ \max}, \qquad i = 1, 2, \ldots, m \end{cases} \tag{1}$$

where X is the m-dimensional matrix of all parameters to be estimated, m is the total number of parameters to be estimated, and x_i represents the parameters to be estimated. $x_{i\ \min}$ and $x_{i\ \max}$ are the minimum

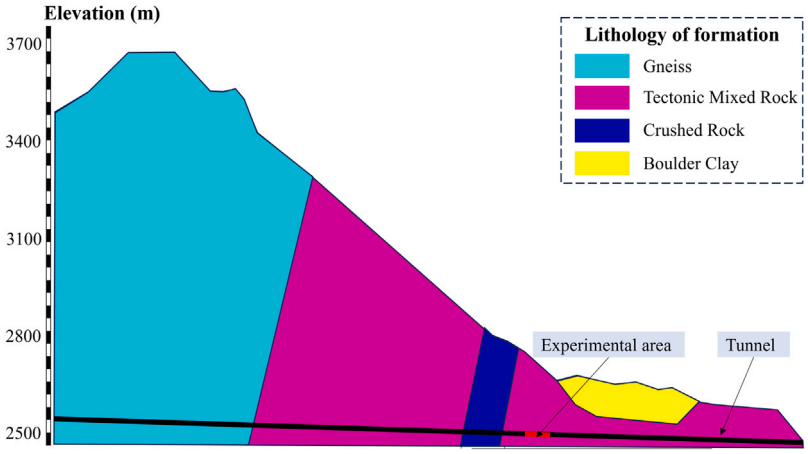


Fig. 4. Study area.

and maximum values of tunnel environmental parameters, determined through laboratory analysis or rock mass classification.

When the number of parameters to be estimated is large, rational experimental design is necessary. In this study, uniform experimental design (UED) is adopted Fang (2003). UED uniformly distributes across the entire parameter space, effectively capturing the interplay between response variables (like displacement, seepage, and temperature) and input variables. This thorough coverage significantly enhances the precision of model training and prediction (Zhai et al., 2019). The UED-derived experimental scheme, XS, is defined as:

$$\begin{cases} XS = \{X_1, X_2, \ldots, X_N\} \\ N = \prod_{i=1}^{m} n_i \end{cases} \tag{2}$$

where X_i represents the ith parameter set N is the total number of parameter sets, and n_i represents the number of divisions for the ith parameter, with the number of divisions determining the number of intervals each variable is uniformly divided into.

Complex rock behaviors such as swelling, fracturing, water inrush, and high geothermal phenomena may manifest as the tunnel face progresses, altering the stress, seepage, and temperature fields in the surrounding rock (Ding et al., 2022; Zhao et al., 2023). Assuming these complex rock behaviors can be simulated numerically, the response values R (encompassing aspects like displacement, water inflow, and temperature in the monitoring area) can be computed based on XS:

$$R = SL(X_i) = [r_{i1}, r_{i2}, \ldots, r_{il}]^T \tag{3}$$

where $r_{i1}, r_{i2}, \ldots, r_{il}$ is the response value of the monitoring area calculated for the ith group of parameters; l represents the number of monitoring data; SL is a mapping function representing the finite element calculation process.

Based on the aforementioned calculations, we generate a sample database $D = \{(X_1, R_1), (X_2, R_2), \ldots, (X_N, R_N)\}$, randomly selecting 80% of it as the training set and the remainder as the test set. This division ratio has been proven to be an appropriate ratio for model training and testing (Kassa and Wubineh, 2023). A GP model is then trained using the sample database D to fit the nonlinear functional relationship between tunnel environmental parameters and response values, represented as f.

The monitoring data within the tunnel, including the settlement of vault, water inflow and internal rock temperature, serve as the foundational data for back-analysis. This data can be represented as $\hat{P} = [\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_l,]^T$, a vector composed of l monitoring data items. Finally, by integrating real monitoring data, this study employs the DE algorithm to search the optimal tunnel environmental parameters. The initial population is iterated through mutation, crossover, and selection operations until the maximum number of iterations is reached or the objective function meets the predetermined value, thus identifying the parameter set X_0 that satisfies the conditions:

$$\begin{cases} J = \left(f(X) - \hat{P}\right)^2 \\ X_0 = \arg\min_X J \end{cases} \tag{4}$$

ii. Simulation based on optimal parameters.

Upon identifying the optimal tunnel environment parameter X_0, this parameter was entered into the numerical model to evaluate the distributions of temperature, stress, and seepage fields around the existing tunnel, thus significantly improving the DT model of the tunnel environment. Furthermore, this method permits the prediction of water inflow at diverse distances, temperature at the tunnel face, and the distribution of each physical field across simulations of tunnel excavation.

5. Experiments and analysis

5.1. Study area and dataset

A section of a mountain tunnel, known for its complexity and challenging conditions, serves as the experimental site for this study. Fig. 4 displays the geological longitudinal section of the area, showing a tunnel that spans 3 km with a depth exceeding 1200 m. The lithological composition mainly consists of schist, with occurrences of granite and various structural rocks. The site is characterized by well-developed joints and rock masses that are predominantly intact yet partly fractured, situated within the Milin fault zone. This fault zone is notable for its intense hydrothermal activity; atmospheric precipitation and melting snow absorb underground heat, leading to the formation of high-temperature thermal water that permeates through rock fractures. As a result, the tunnel environment experiences high geothermal temperatures, significant seepage pressures, and strong geostresses, frequently causing high-temperature water inrush incidents. The complex tunnel environment, combined with the limitations of detection methods and the multi-source, heterogeneous, and incomplete nature of the available detection data, presents substantial challenges in accurately developing a DT model of the tunnel's geological environment using conventional approaches. Therefore, this particular tunnel was selected for the experimental demonstration of the methodologies proposed in this study.

The experimental data, depicted in Fig. 5, are sourced from actual engineering site detection and include:

(1) Tunnel Design Model: Mainly consisting of the tunnel BIM and design parameters, these are employed to model the tunnel's geometric configuration.

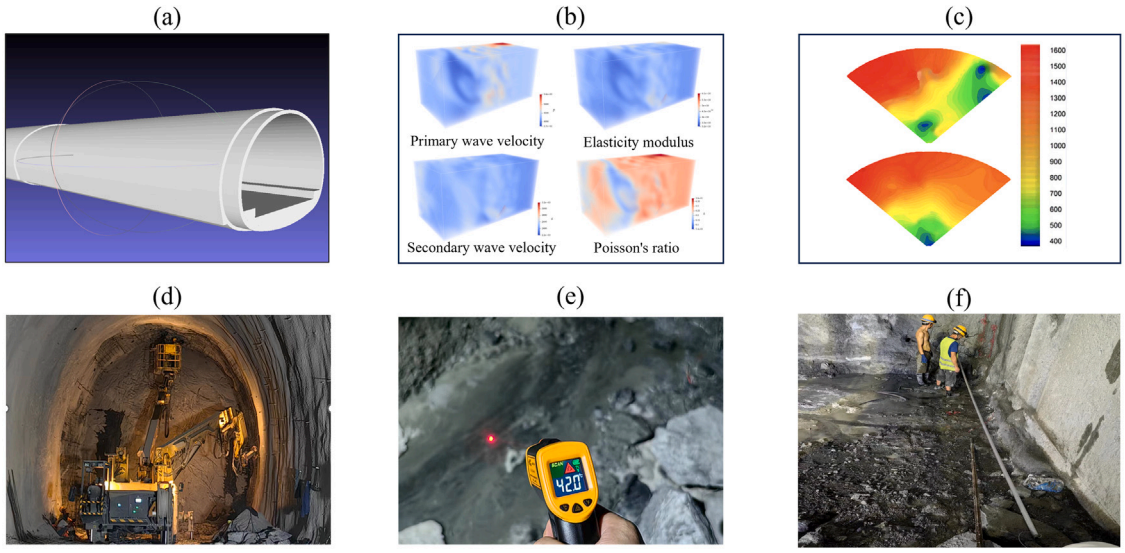


Fig. 5. Multi-source data. (a) Tunnel design BIM model, (b) TSP data, (c) TEM data, (d) Tunnel vault settlement, (e) Temperature of rock, (f) Water inflow.

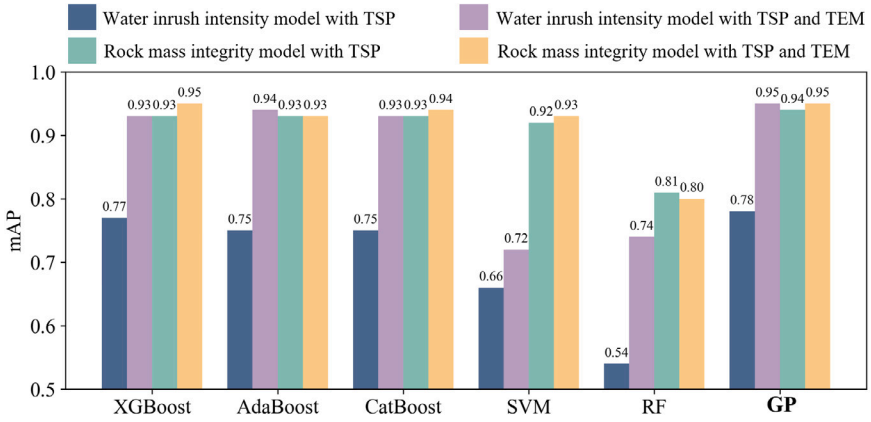


Fig. 6. Comparison of performance of different machine learning models.

(2) Advanced geological forecasting data: Comprising TSP data (such as longitudinal and transverse wave velocities, Poisson's ratio, Young's modulus, and rock density), TEM data (resistivity), information from advanced horizontal drilling, and geophysical interpretation reports, these data are instrumental in the 3D model inversion of water-rich fractured zones.

(3) On-Site Monitoring Data: Including measurements like surface and internal temperatures of the rock at the excavation face, tunnel wall rock temperature, and the temperature and water inflow at the excavation face, as well as tunnel vault settlement, these data points are crucial for the back-analysis of parameters.

5.2. DT modeling and accuracy evaluation

Due to the variability of tunnel geological conditions, a significant imbalance exists in the collected sample categories related to water inrush intensity and rock integrity. Maintaining a data sample ratio of approximately 1:1 is essential to more accurately reflect classification results (Ruben et al., 2022). Consequently, this study employed the adaptive synthetic algorithm to address the imbalance in sample data.

Given the smaller data range of TEM compared to TSP, and the existence of spatial areas with only TSP data, predictive models were constructed using either TSP data alone or a combination of TSP and TEM data. To ensure the precision of the DT model, this study extensively evaluated algorithms including XGBoost, AdaBoost, CatBoost, SVM, RF, and GP. The GP model employed maximum likelihood estimation for hyperparameter optimization, while Bayesian optimization algorithms were applied to the other models. The experimental results, presented in Fig. 6, utilized mAP as the criterion for comparative analysis of the six algorithms. The results indicate that for evaluating rock mass integrity, XGBoost, AdaBoost, CatBoost, SVM, and GP all demonstrated high performance, with mAP values exceeding 0.92, significantly outperforming RF, which achieved mAP values of 0.81 and 0.80. Notably, the simultaneous use of TEM and TSP data did not significantly enhance the performance of any model. In the assessment of water inrush intensity, models using only TSP data yielded lower mAP values, ranging from 0.54 to 0.78. However, the introduction of TEM data significantly improved the performance of all models, with XGBoost, AdaBoost, and CatBoost achieving mAP values greater than 0.93, notably outperforming SVM at 0.72 and RF at 0.74. In summary, while TSP data alone is effective for evaluating rock mass integrity, it is insufficient for accurately assessing water inrush intensity. The inclusion of TEM data, however, significantly enhances the accuracy of water inrush intensity evaluations. XGBoost, AdaBoost, CatBoost, and GP all demonstrated strong performance on the dataset used in this study, making them viable options for subsequent geological modeling. The GP model was selected for DT modeling due to its excellent performance in our experiments, particularly its effectiveness in handling small, noisy datasets. In complex mountainous tunnel environments, where data collection is challenging and often results in incomplete,

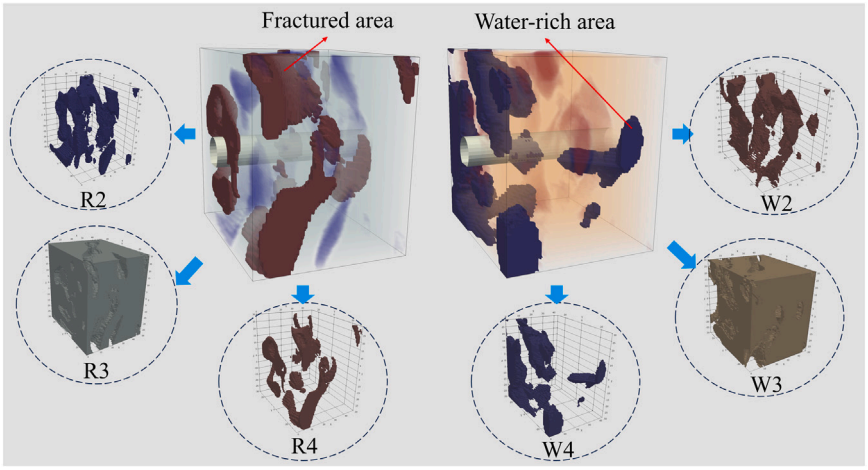


Fig. 7. Three-dimensional distribution of rock mass integrity and water inrush intensity.

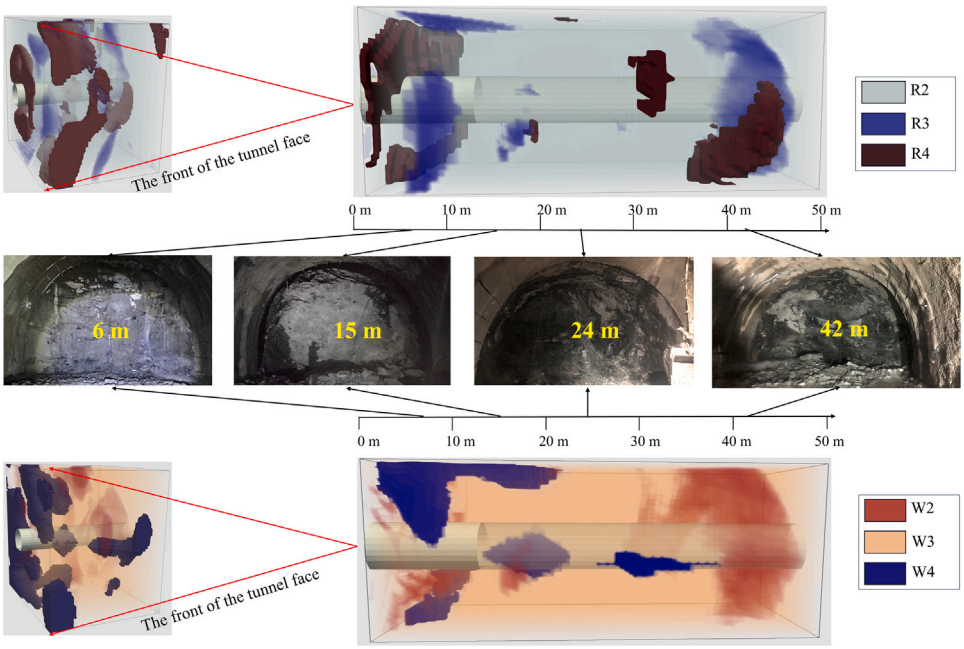


Fig. 8. Reliability verification of three-dimensional model of rock mass integrity and water inrush intensity.

high-noise data with small sample sizes and significant measurement errors, the GP model demonstrated exceptional adaptability (Liu et al., 2020). Unlike algorithms such as XGBoost and CatBoost, GP, grounded in Bayesian theory, is trained using the structural risk minimization principle, making it ideal for small-sample learning. Additionally, GP's hyperparameters can be optimized adaptively (Su et al., 2017).

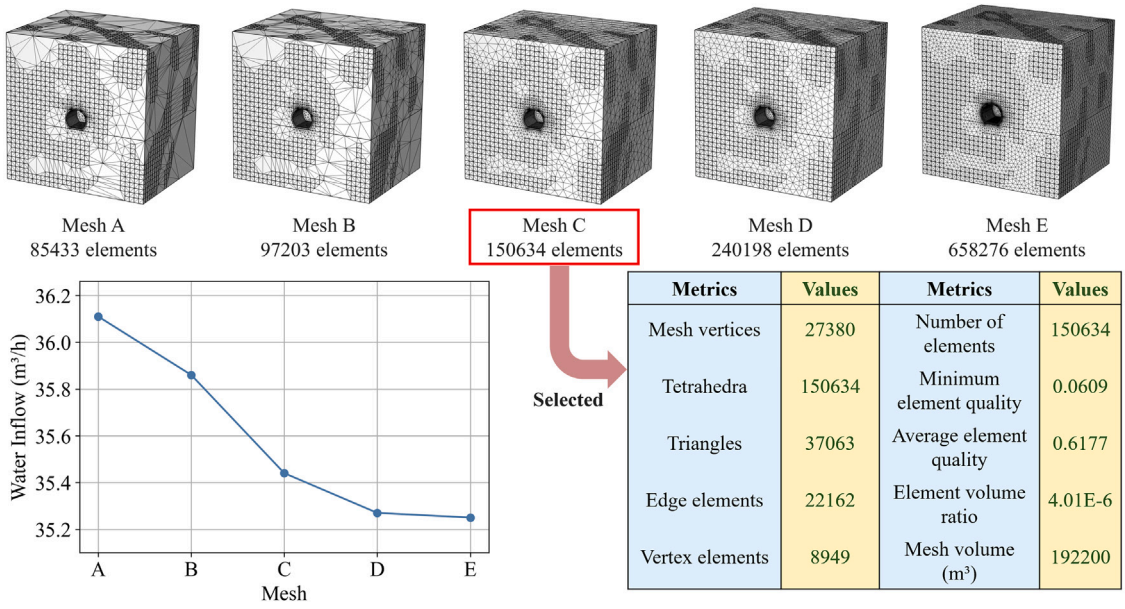
Subsequent to model training completion, the developed model was applied to an actual engineering project, focusing on predicting the geological environment surrounding the tunnel. A specific excavation section of the tunnel served as the experimental location. Advanced geological exploration at the tunnel face was conducted, followed by data preprocessing, after which the trained GP model was used for predicting the tunnel's surrounding rock. The results yielded a spatial three-dimensional distribution of rock integrity and water inrush intensity within a 50 m (width) × 60 m (length) × 60 m (height) range, as illustrated in Fig. 7. These findings revealed a predominantly relatively fractured rock composition along the tunnel's centerline, with the water-rich area situated towards the right front of the tunnel face, corroborating the comprehensive interpretation report from the design institute.

To confirm the accuracy of the DT model, which depicts rock mass integrity and water inrush intensity. We juxtapose the tunnel face at distances of 6 m, 15 m, 24 m, and 42 m ahead of the excavation with the corresponding sections of the DT model. Fig. 8 reveals partly broken rock masses at 6 m, 15 m, and 24 m from the tunnel face, and a partly intact rock mass at 42 m, aligning with the DT model. Regarding water inrush intensity, the areas 6 m and 15 m in front of the tunnel face are identified as gush water inrush zones, whereas the areas 24 m and 42 m are classified as linear water inrush zones, aligning with the DT model. This correspondence underlines the three-dimensional model's high reliability in assessing rock mass integrity and water inrush strength, thus providing a dependable reference for tunnel construction planning.

5.3. Parameter calibration and simulation evaluation

5.3.1. Numerical model

Based on the tunnel design parameters, rock integrity, and three-dimensional model of water inrush intensity, a finite element analysis geometric model was constructed with dimensions of 50 m (width) ×



Metrics	Values	Metrics	Values
Mesh vertices	27380	Number of elements	150634
Tetrahedra	150634	Minimum element quality	0.0609
Triangles	37063	Average element quality	0.6177
Edge elements	22162	Element volume ratio	4.01E-6
Vertex elements	8949	Mesh volume (m³)	192200

Fig. 9. Mesh sensitivity analysis.

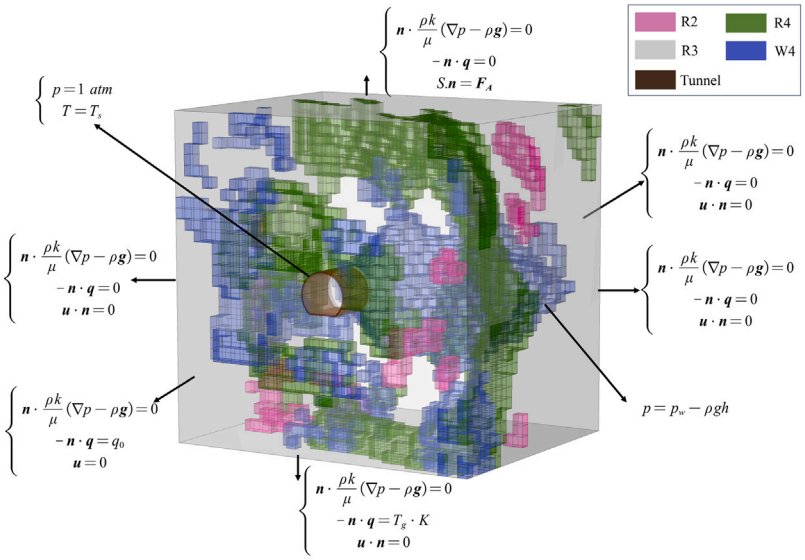


Fig. 10. Approximate geometric model and boundary conditions.

60 m (length) × 60 m (height). Due to the complexity of water-rich areas and broken zones, numerous intersecting and overlapping regions in the space made mesh division difficult. Setting a smaller mesh size could easily lead to computational non-convergence. To simplify the calculation, an approximate geometric shape for the three-dimensional model of rock integrity and water inrush intensity was used, substituting the original model's complex geometry with 2m-sided cubes. This strategic simplification aims to enhance computational efficiency while preserving essential features of the geological environment, thereby facilitating a more manageable and effective finite element analysis.

In this study, we employed free tetrahedral elements for mesh generation, with local mesh refinement applied around the tunnel lining and surrounding rock, while reducing mesh density in regions farther from the tunnel. This approach effectively balances the trade-off between accuracy and computational cost. To ensure that the selected mesh resolution does not significantly affect the accuracy of results, we conducted a sensitivity analysis by comparing water inflow data across five different mesh resolutions, as illustrated in Fig. 9. Based on considerations of computational speed and efficiency, Mesh C was chosen as the standard for subsequent mesh refinement.

The approximate geometric model and boundary conditions are shown in Fig. 10.

Solid Mechanics Boundary: The front, back, left, and right boundaries of the model are roller support constraints. Based on geological exploration data, the tunnel burial depth is 230 m. According to the "Code for Design of Highway Tunnel" (JTG D70-2004) (Standard, 2004), the uniform load applied to the upper boundary F_A is calculated as $F_A = \gamma \cdot H$, where γ represents the unit weight of the rock above the tunnel (taken as 25.5 kN/m^3 based on the rock density), and H is the burial depth of the upper boundary of the model (200 m). This calculation results in $F_A = 5.1 \times 10^6$ N/m^2. The lower boundary has fixed displacement constraints, while the tunnel face and the rock walls inside the tunnel are modeled as free boundaries.

Hydraulic boundary based on Darcy's law: The front, back, left, right, and upper and lower boundaries of the model are set as impermeable boundaries. Assuming that the direction of the hydraulic gradient

Table 1
Range of tunnel environmental parameters.

Tunnel environmental parameters	Partly intact rock (R2)	Partly broken rock (R3)	Broken rock (R4)
Density ρ (kg/m3)	2700	2500	2300
Elastic modulus E (GPa)	[21~31]	[11~21]	[1~11]
Poisson's ratio μ	0.23	0.28	0.32
Internal friction angle Φ (°)	50	40	33
Cohesion λ (MPa)	[1.1~1.5]	[0.6~1.1]	[0.2~0.6]
Porosity Po (%)	[5~15]	[15~25]	[25~35]
Permeability Pe (m^2)	[10^{-15}~10^{-14}]	[10^{-14}~10^{-13}]	[10^{-13}~10^{-12}]
Pressure of rich water area P_w (MPa)	[0.1~2]		
Thermal conductivity Tc (W/(m K))	[3.7~4.2]	[3.2~3.7]	[2.7~3.2]
Specific heat capacity C (J/(kg K))	850		
Geothermal gradient Tg (°C/m)	[0.01~0.5]		

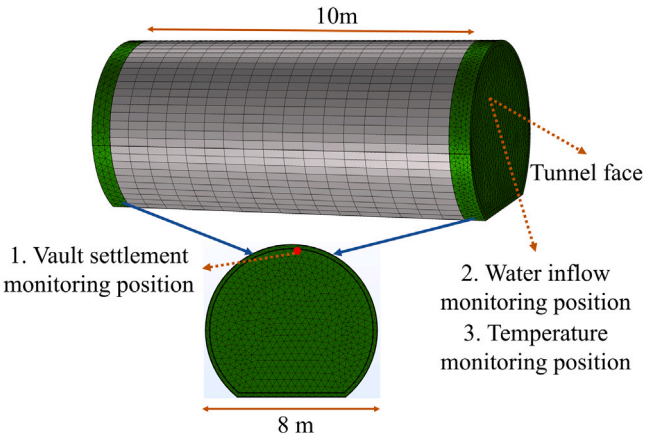


Fig. 11. Tunnel monitoring position.

in the region is consistent with gravity, only the water pressure in the W4 region, namely the rich water area, is considered here, and the water pressure is set as $p = p_w - \rho g h$, where p_w is a fixed pressure in Pa, ρ is the water density in kg/m^3, g is the gravitational acceleration in m/s^2, and h is the depth in meters. The pressure on the tunnel face and the rock walls around the tunnel is set to 1 atmospheric pressure.

Porous media heat transfer boundary: The front, back, left, right, and upper boundaries of the model are set as thermal insulators. Assuming the geothermal gradient direction in the region is consistent with gravity, the geothermal flux on the lower boundary of the model is $Flux = T_g \cdot K$, where T_g is the geothermal gradient in ℃/m and K is the average thermal conductivity of the lower boundary in W/(m·℃). Since the rock wall temperature will not reach a steady-state equilibrium with the air temperature shortly after blasting at the tunnel face, the boundary temperature of the tunnel face and rock walls inside the tunnel is set to a fixed temperature $T = T_s$, where T_s is the rock wall surface temperature.

The monitoring area designated for the tunnel, illustrated in Fig. 11, features a monitoring point atop each tunnel section. A monitoring section is set up at 10-meter intervals to document vault settlement, integrating two displacement monitoring sections into the model. Moreover, a monitoring surface is strategically placed at the tunnel face to collect data on water inflow and the average surface temperature at the tunnel face.

5.3.2. Back-analysis

The range of values for environmental parameters within the tunnel is detailed in Table 1. The model applies elastoplastic theory and the Mohr–Coulomb failure criteria to define mechanical parameters. Elastic modulus and cohesion, identified as irreversible variables in Liu's research (Li et al., 2023), are thus earmarked for back-analysis. Porosity, permeability, and hydraulic gradient, vital for influencing groundwater flow and the stability of subterranean structures, are deemed critical hydraulic parameters and selected for back-analysis. In the context of thermal analysis, the geothermal gradient and thermal conductivity, pivotal for simulating geothermal flow, are chosen as parameters for back-analysis. The determination of these parameter value ranges is based on laboratory analyses and extensive references from the Engineering Geology Handbook (The Compilation Committee of Engineering Geology Handbook, 2007) and "Code for Geological Investigation of Water Resources and Hydropower Engineering" (GB50487, 2008), ensuring a thorough and comprehensive approach to accurately modeling the tunnel environmental conditions.

Due to the large number of variables, UED is employed. A 30-level uniform experimental design table is selected, designing 30 combinations of 17 parameters, and the parameters of each scheme are incorporated into the mechanical-hydraulic-thermal multi-field coupled model for finite element calculations. This yields values for the settlement of vault, water inflow, and tunnel face surface temperature. The grey relational grade (GRG) is used to assess the correlation between input and output parameters, hence selecting sensitive parameters. The calculation formula is as follows:

$$\begin{cases} \xi_{ij} = \dfrac{\Delta_{\min} + \rho \cdot \Delta_{\max}}{\Delta_{ij} + \rho \cdot \Delta_{\max}}, \\ CRG_j = \dfrac{1}{n} \sum_{i=1}^{n} \xi_{ij} \end{cases} \tag{5}$$

where ξ_{ij} is the grey relational coefficient, Δ_{ij} is the difference between the reference and comparison sequences, $\Delta_{\min}$ and $\Delta_{\max}$ are the minimum and maximum values of all differences, respectively. ρ is the resolution coefficient, usually set to 0.5. CRG_j is the relational grade of the jth variable, and n is the number of sample points.

The calculation results, as shown in Fig. 12, P_1, P_2, P_3, P_4 represent two monitored tunnel vault, water inflow, and internal rock temperature. When the GRG value between input and output parameters is greater than or equal to 0.67, the input parameter is deemed sensitive; otherwise, it is considered insensitive. To simplify the back-analysis process, only sensitive parameters E_{R2}, E_{R3}, Po_{R3}, Pe_{R3}, Tc_{R4}, P_w, T_g are retained for back-analysis, while the remaining non-sensitive parameters were set to the midpoints of their respective ranges. This approach enhances both the efficiency and accuracy of the back-analysis.

After identifying seven pivotal parameters for back-analysis, this study consulted further engineering data to determine the remaining parameter values. Employing UED methodology, sample datasets were meticulously generated through finite element forward computations. To tackle the challenges associated with performance discrepancies in small sample datasets, this research employed GP, SVM, XGBoost, and RF models. These models were instrumental in establishing complex, nonlinear relationships between input factors and corresponding outputs. The results, as shown in Fig. 13(a), indicate that the GP model performs best, slightly better than SVM, with R-square (R2) of 0.96, mean absolute percentage error (MAPE) of 4.94%, and root mean square error (RMSE) of 0.47, while XGBoost and RF perform poorly on

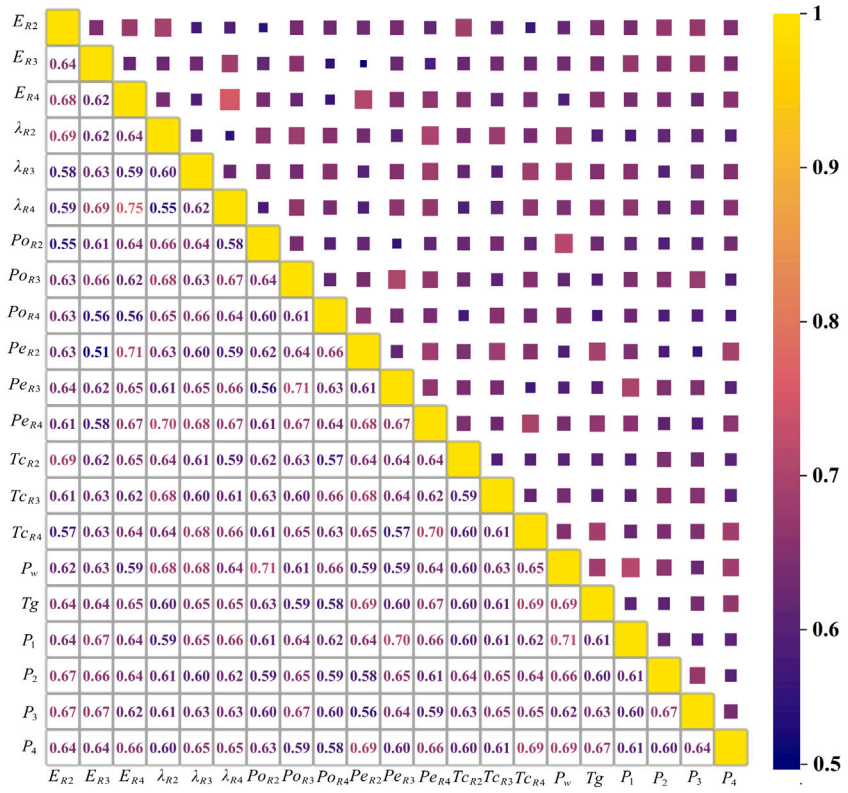


Fig. 12. CRG values among input and output variables.

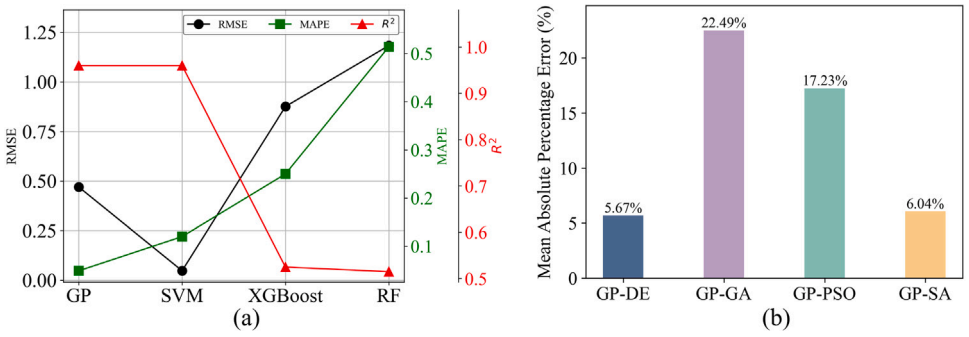


Fig. 13. (a) Comparison of performance indicators of different machine learning models, (b) Performance comparison of different optimization algorithms.

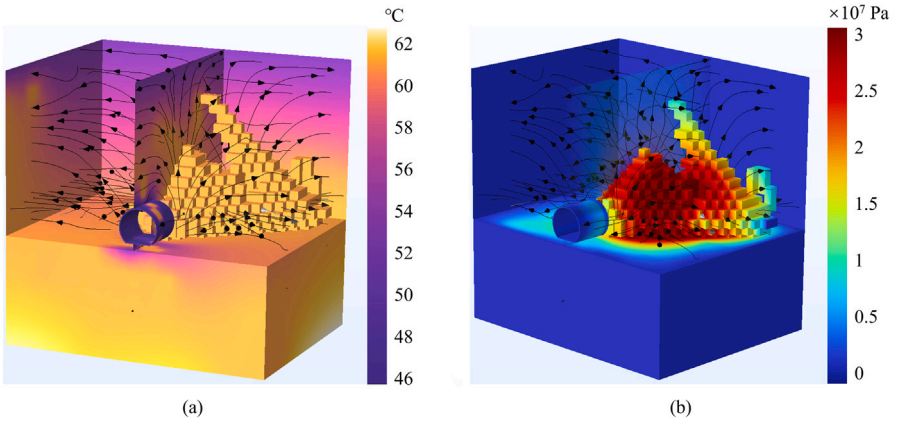


Fig. 14. (a) Distribution of temperature field in tunnel, (b) distribution seepage field in tunnel.

Table 2
Comparison of measured values and calculated values based on optimal parameters.

Parameters	Results of back-analysis	Field measured values	Relative errors (%)
Tunnel vault settlement1 (mm)	15.6	15.0	3.6
Tunnel vault settlement2 (mm)	13.7	15.4	9.0
Water inflow (m^3/h)	35.4	33.0	7.2
Internal rock temperature (°C)	67.1	67.5	0.6

Table 3
Reliability verification of DT model of tunnel geological environment.

Parameters	Excavate 6 m		Excavate 15 m		MAPE (%)
	Predicted value	Monitored value	Predicted value	Monitored value	
Tunnel vault settlement (mm)	15.1	14.5	15.2	15.1	2.4
Water inflow (m^3/h)	35.3	33.0	18.2	23.5	14.8
Internal rock temperature (°C)	70.1	68.1	69.5	67.8	2.7

the current dataset. Incorporating actual tunnel monitoring data, this study employs the GP model for back-analysis of parameters, utilizing various optimization algorithms. The results are shown in Table 2. The effectiveness of GP-DE, GP-GA, GP-PSO, and GP-SA algorithms is assessed by comparing the finite element results derived from their optimal parameter combinations against the monitoring data, as depicted in Fig. 13(b). The DE algorithm outperforms the others, especially SA, while GA and PSO appear to be limited by local optima, leading to reduced accuracy. The optimal tunnel environmental parameter group is: E_{R2} = 12.91 GPa, E_{R3} = 3.52 GPa, Po_{R3} = 0.23, Pe_{R3} = 9.0×10^{-14} m^2, Tc_{R4} = 3.05 W/(m ℃), P_w = 1.25 MPa, T_g = 0.13 ℃/m.

Inputting the optimal tunnel environmental parameters into the finite element model, the results as shown in Table 2 indicate relative errors of 3.6%, 9.0%, 7.2%, and 0.6% for the four monitored variables, with MAPE of 5.1%. This demonstrates that the method proposed in this study can effectively invert tunnel environmental parameters.

5.3.3. Simulation based on optimal parameters

Optimal parameters were applied for simulation, as depicted in Fig. 14. The analysis of the temperature field revealed a gradual increase in temperature from the surface to deeper underground levels. However, temperatures near the tunnel walls remained lower, illustrating the thermal conductivity characteristics of the subsurface environment and the cooling effects induced by tunneling activities. Regarding the seepage field, the findings indicate that water pressure peaks within water-rich areas, subsequently dispersing towards the tunnel's palm surface and adjacent areas of lower pressure. This behavior aligns with established theories of water movement within porous media. Integrating these multifaceted simulation outcomes into a voxel-based DT framework significantly broadens the model's applicability and depth. Moreover, this integration provides extensive data support, crucial for the design and execution phases of tunnel construction projects.

5.3.4. Reliability evaluation

To verify the accuracy of the proposed method, predictions regarding the vault settlement, surrounding rock temperature, and water inflow at positions 6 m and 15 m ahead of the excavation face were made. These predictions were then compared with on-site monitoring data, as detailed in Table 3. The forecasts for vault settlement and internal rock temperature were highly accurate, exhibiting a MAPE of 5.4% and 2.9%, respectively. However, the accuracy of water inflow predictions at the tunnel face was comparatively lower, at 16.1%. Field observations indicate that the tunnel water inflow at the face is significantly influenced by the location and quantity of deepened boreholes. Consequently, variability in predicted water inflow is anticipated, given the non-uniform distribution of surrounding rock fractures.

The results suggest that the excavation analysis, based on the DT model, closely aligns with the subsequent construction excavation process. The DT model of the tunnel's geological environment, constructed in this study, accurately depicts the actual distribution of water-rich and fractured zones in the tunnel, showcasing high accuracy and reliability.

5.4. Discussion

High-temperature water inrush poses significant risks to complex mountainous tunnel construction. Accurate geological modeling and reliable simulation analysis are crucial for disaster prediction and the optimization of construction design. This paper distinguishes itself from previous studies on water inrush disaster simulation in several key aspects:

(1) This study improves the efficiency of tunnel engineering. By leveraging a digital twin framework, we integrated multi-source heterogeneous data, multidisciplinary mechanistic models, and high-performance algorithms to achieve automated and precise geological modeling as well as reliable simulation of water inrush hazards. In practical applications, the entire process of data acquisition, processing, analysis, and visualization is completed in approximately 2 h. In contrast, according to feedback from on-site engineers, traditional rock mass environment prediction typically takes around 2 days, involving manual sketching of the tunnel face, processing of advance geological prediction data, followed by interpretation by specialized engineers, and finally sending back a comprehensive assessment report to guide construction (Shi et al., 2017; Liu and Gan, 2023; Liu et al., 2009; Huang et al., 2022). This approach not only significantly improves efficiency but also ensures accuracy. Compared to the uncertainties that may arise from manual interpretation, the results obtained through high-performance machine learning algorithms and coupled simulations in this study are far more reliable.

(2) An innovative approach is introduced by grading two key factors that influence water inrush: rock mass integrity and water inflow intensity. These classifications are then used to develop a high-resolution digital twin geological model, significantly enhancing the reliability of simulations. A key aspect of this research is the integration of high-resolution advanced geological forecasting data, which has enabled the construction of a geological model with a resolution of 1 meter—sufficient to support high-precision water inrush simulations. In contrast, previous studies typically relied on geological models derived from interpolated sparse borehole data, whose resolution is far inferior to that of models obtained directly from geophysical data at the tunnel face (Liu et al., 2022; Xie et al., 2023; Wang et al., 2024b; Gao et al., 2019). The inaccuracies in these lower-resolution geological models and their boundary conditions have often compromised the reliability of simulation.

(3) This paper presents a novel GP-DE back-analysis parameter calibration method developed within a THM coupled model. By utilizing temperature, water inflow, and displacement monitoring data from within the tunnel, this method directly or indirectly calibrates the key simulation parameters, thereby further enhancing the reliability

of the simulations. In contrast, previous studies have typically relied on displacement observations alone for back-analysis to calibrate rock mass parameters (Zhang et al., 2020; Deng et al., 2022; Li et al., 2023), an approach that is evidently insufficient for the complexities of a THM-coupled tunnel environment. Additionally, many studies have depended on empirical values for numerical simulation parameters. Due to the inherent differences between geological environments, these empirical values often lack broad applicability, leading to relatively lower reliability in the simulation results.

At the same time, the method proposed in this paper also has the following limitations:

(1) The 3D back-analysis accuracy for the water-rich broken zone heavily depends on the comprehensiveness of the sample database. Challenges in data collection and a lack of diversity in the dataset limit the model's ability to cover all complex geological conditions, affecting its applicability and precision.

(2) The employed mathematical model has been simplified to suit the tunnel environment, which may not sufficiently depict high-temperature inrush water scenarios in more complex geological settings.

6. Conclusions

This study introduces a novel method for simulating high-temperature water inrush incidents based on a DT model of the tunnel's geological environment, aimed at tackling water inrush and high temperatures in complex tunnel construction. The approach integrates voxel models with advanced geological prediction data, utilizing high-performance machine learning algorithms to construct a DT model of rock mass integrity and water inrush intensity. By combining machine learning algorithms with the finite element method and employing GP-DE for multi-source tunnel data joint back-analysis, the study accurately estimates optimal tunnel environmental parameters, facilitating reliable simulation analysis of high-temperature water inrush. Overall, this methodology provides a novel approach to addressing complex challenges in tunnel engineering and establishes a solid foundation for future research. Future developments will build on this study, further enhancing digital twin models and simulation methods to more effectively mitigate various disaster risks in tunnel engineering.

CRediT authorship contribution statement

Yang He: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Yulin Ding:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Qing Zhu:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Haoyu Wu:** Visualization, Formal analysis, Conceptualization. **Yongxin Guo:** Project administration, Data curation. **Qiang Wang:** Investigation, Formal analysis, Data curation. **Runfang Zhou:** Resources, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

Aleksandrov, M., Zlatanova, S., Heslop, D.J., 2021. Voxelisation algorithms and data structures: A review. Sensors 21 (24), http://dx.doi.org/10.3390/s21248241, URL: https://www.mdpi.com/1424-8220/21/24/8241.

Berti, M., Bertello, L., Bernardi, A.R., Caputo, G., 2017. Back analysis of a large landslide in a flysch rock mass. Landslides 14 (6), 2041–2058. http://dx.doi.org/10.1007/s10346-017-0852-5.

Chang, X., Wang, H., Zhang, Y., 2023. Back analysis of rock mass parameters in tunnel engineering using machine learning techniques. Comput. Geotech. 163, http://dx.doi.org/10.1016/j.compgeo.2023.105738.

Chen, Y., 2021. Back analysis and numerical simulation of surrounding rock parameters of tunnel close construction based on PSO-BP algorithm. In: Huang, C., Chan, Y.-W., Yen, N. (Eds.), 2020 International Conference on Data Processing Techniques and Applications for Cyber-Physical Systems. Springer Singapore, Singapore, pp. 1237–1243.

Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system. In: KDD'16: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. Assoc Comp Machinery; Assoc Comp Machinery SIGMOD; Assoc Comp Machinery SIGKDD, pp. 785–794. http://dx.doi.org/10.1145/2939672.2939785, 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), San Francisco, CA, AUG 13-17, 2016.

Cui, S., Zhang, B., Feng, F., Xie, L., 2015. The research of geological forecast based on muti-source information fusion. In: Lollino, G., Giordan, D., Thuro, K., Carranza-Torres, C., Wu, F., Marinos, P., Delgado, C. (Eds.), Engineering Geology for Society and Territory - Volume 6. Springer International Publishing, Cham, pp. 1001–1005.

Deng, H.-S., Fu, H.-L., Shi, Y., Zhao, Y.-Y., Hou, W.-Z., 2022. Countermeasures against large deformation of deep-buried soft rock tunnels in areas with high geostress: A case study. Tunn. Undergr. Space Technol. 119, 104238. http://dx.doi.org/10.1016/j.tust.2021.104238.

Ding, Q., Li, B., Su, H., Xu, N., Li, X., Deng, X., 2022. Damage mechanism and stability analysis of rock mass in the high geo-stress tunnel subjected to excavation. Geomat. Nat. Hazards Risk 13 (1), 75–93. http://dx.doi.org/10.1080/19475705.2021.2015458.

Dong, L.-j., Li, X.-b., Peng, K., 2013. Prediction of rockburst classification using random forest. Trans. Nonferrous Metals Soc. China 23 (2), 472–477. http://dx.doi.org/10.1016/S1003-6326(13)62487-5.

El Saddik, A., Laamarti, F., Alja Afreh, M., 2021. The potential of digital twins. IEEE Instrum. Meas. Mag. 24 (3), 36–41.

Erharter, G.H., Weil, J., Tschuchnigg, F., Marcher, T., 2022. Potential applications of machine learning for BIM in tunnelling. Geomech. Tunn. 15 (2), 216–221.

Fan, H., Li, L., Chen, G., Liu, H., Gao, J., Li, C., Peng, X., Zhou, S., 2023a. Analysis method of the water inrush and collapse in jointed rock mass tunnels: A case study. Eng. Anal. Bound. Elem. 146, 838–850. http://dx.doi.org/10.1016/j.enganabound.2022.11.030.

Fan, H., Li, L., Liu, H., Shi, S., Hu, J., Zhou, S., 2021. Advanced stability analysis of the tunnels in jointed rock mass based on TSP and DEM. KSCE J. Civ. Eng. 25 (4), 1491–1503. http://dx.doi.org/10.1007/s12205-021-0170-2.

Fan, H., Li, L., Zong, P., Liu, H., Yang, L., Wang, J., Yan, P., Sun, S., 2023b. Advanced stability analysis method for the tunnel face in jointed rock mass based on DFN-DEM. Undergr. Space 13, 136–149. http://dx.doi.org/10.1016/j.undsp.2023.03.009.

Fang, K., 2003. Theory, method and applications of the uniform design. In: Pham, H., Lu, M. (Eds.), Eighth ISSAT International Conference on Reliability and Quality in Design, Proceedings. ISSAT, pp. 235–239, 8th ISSAT Conference on Reliability and Quality in Design, Anaheim, CA, AUG 07-09, 2002.

Feng, X., Zhang, Z., Sheng, Q., 2000. Estimating mechanical rock mass parameters relating to the three gorges project permanent shiplock using an intelligent displacement back analysis method. Int. J. Rock Mech. Min. Sci. 37 (7), 1039–1054. http://dx.doi.org/10.1016/S1365-1609(00)00035-6.

Gangrade, R., Grasmick, J., Trainor-Guitton, W., Mooney, M., 2022a. Risk-based methodology to optimize geotechnical site investigations in tunnel projects. Tunn. Undergr. Space Technol. 127, 104589. http://dx.doi.org/10.1016/j.tust.2022.104589.

Gangrade, R.M., Mooney, M.A., Trainor-Guitton, W., 2022b. Quantification of stratigraphic transition location uncertainty for tunneling projects. J. Geotech. Geoenviron. Eng. 148 (8), 04022057. http://dx.doi.org/10.1061/(ASCE)GT.1943-5606.0002815.

Gao, C., Li, L., Zhou, Z., Li, Z., Cheng, S., Wang, L., Zhang, D., 2021. Peridynamics simulation of water inrush channels evolution process due to rock mass progressive failure in karst tunnels. Int. J. Geomech. 21 (4), http://dx.doi.org/10.1061/(ASCE)GM.1943-5622.0001980.

Gao, C.-l., Zhou, Z.-q., Yang, W.-m., Lin, C.-j., Lia, L.-p., Wang, J., 2019. Model test and numerical simulation research of water leakage in operating tunnels passing through intersecting faults. Tunn. Undergr. Space Technol. 94, http://dx.doi.org/10.1016/j.tust.2019.103134.

GB50487, 2008. Code for geological investigation of water resources and hydropower engineering.

Gottron, D., Henk, A., 2021. Upscaling of fractured rock mass properties - an example comparing discrete fracture network (DFN) modeling and empirical relations based on engineering rock mass classifications. Eng. Geol. 294, http://dx.doi.org/10.1016/j.enggeo.2021.106382.

Gutierrez, M., 2020. Learning from field measurements through heuristic back analysis. In: Correia, A., Tinoco, J., Cortez, P., Lamas, L. (Eds.), Information Technology in Geo-Engineering. In: Springer Series in Geomechanics and Geoengineering, pp. 18–29. http://dx.doi.org/10.1007/978-3-030-32029-4_2, 3rd International Conference on Information Technology in Geo-Engineering (ICITG), Guimaraes, PORTUGAL, SEP 29-OCT 02, 2019.

Hillier, M.J., Schetselaar, E.M., de Kemp, E.A., Perron, G., 2014. Three-dimensional modelling of geological surfaces using generalized interpolation with radial basis functions. Math. Geosci. 46, 931–953.

Hu, W., Zhang, T., Deng, X., Liu, Z., Tan, J., 2021. Digital twin: A state-of-the-art review of its enabling technologies, applications and challenges. J. Intell. Manuf. Spec. Equip. 2 (1), 1–34.

Huang, M., Ninić, J., Zhang, Q., 2021. BIM, machine learning and computer vision techniques in underground construction: Current status and future perspectives. Tunn. Undergr. Space Technol. 108, 103677. http://dx.doi.org/10.1016/j.tust.2020.103677.

Huang, C., Zhang, S., Wu, S., Gao, Y., 2022. Research and application of a comprehensive forecasting system for tunnels in water-bearing fault fracture zones: A case study. Arab. J. Geosci. 15 (2), 171.

Jian, Z., Xibing, L., Xiuzhi, S., 2012. Long-term prediction model of rockburst in underground openings using heuristic algorithms and support vector machines. Safety Sci. 50 (4, SI), 629–644. http://dx.doi.org/10.1016/j.ssci.2011.08.065, 1st International Symposium on Mine Safety Science and Engineering (ISMSSE), Beijing, PEOPLES R CHINA, OCT 26-29, 2011.

Jørgensen, F., Møller, R.R., Nebel, L., Jensen, N.-P., Christiansen, A.V., Sandersen, P.B., 2013. A method for cognitive 3D geological voxel modelling of AEM data. Bull. Eng. Geol. Environ. 72, 421–432.

Kaiser, P., Zou, D., Lang, P., 1990. Stress determination by back-analysis of excavation-induced stress changes — a case study. Rock Mech. Rock Eng. 23 (3), 185–200. http://dx.doi.org/10.1007/BF01022953.

Kassa, S.M., Wubineh, B.Z., 2023. Use of machine learning to predict california bearing ratio of soils. Adv. Civ. Eng. 2023, http://dx.doi.org/10.1155/2023/8198648.

Khan, M.Y., Xue, G.-Q., Chen, W.-Y., Boateng, C.D., 2020. Investigation of groundwater in-rush zone using petrophysical logs and short-offset transient electromagnetic (SOTEM) data. J. Environ. Eng. Geophys. 25 (3), 433–437. http://dx.doi.org/10.32389/JEEG18-111.

Latif, K., Sharafat, A., Seo, J., 2023. Digital twin-driven framework for TBM performance prediction, visualization, and monitoring through machine learning. Appl. Sci. 13 (20), http://dx.doi.org/10.3390/app132011435.

Li, H., Chen, W., Tan, X., Tan, X., 2023. Back analysis of geomechanical parameters for rock mass under complex geological conditions using a novel algorithm. Tunn. Undergr. Space Technol. 136, http://dx.doi.org/10.1016/j.tust.2023.105099.

Li, J., Heap, A.D., 2014. Spatial interpolation methods applied in the environmental sciences: A review. Environ. Model. Softw. 53, 173–189. http://dx.doi.org/10.1016/j.envsoft.2013.12.008.

Li, L., Jia, C., Sun, Z., Liu, H., Cheng, S., 2021a. Research status and development trend of major engineering disaster prevention and control technology in deep underground. J. Cent. South Univ. (Sci. Technol.) 52 (8), 2539–2556.

Li, S., Liu, B., Sun, H., Nie, L., Zhong, S., Su, M., Li, X., Xu, Z., et al., 2014. State of art and trends of advanced geological prediction in tunnel construction. Chin. J. Rock Mech. Eng. 33 (6), 1090–1113.

Li, J., Wang, Y., Tan, Z., Du, W., Liu, Z., 2021b. Study on water inflow variation law of no.1 shaft auxiliary shaft in HighLiGongshan based on dual medium model. Symmetry 13 (6), http://dx.doi.org/10.3390/sym13060930.

Liu, M., Gan, Q., 2023. Applied research of comprehensive advance geological prediction in Daluoshan water diversion tunnel. Sci. Rep. 13 (1), http://dx.doi.org/10.1038/s41598-023-36090-8.

Liu, H., Li, W., Gu, S., Cheng, L., Wang, Y., Xu, J., 2023a. Three-dimensional modeling of fault geological structure using generalized triangular prism element reconstruction. Bull. Eng. Geol. Environ. 82 (4), http://dx.doi.org/10.1007/s10064-023-03166-8.

Liu, B., Li, S., Zhang, Q., Xue, Y., 2009. Study of the prediction of karstfractured groundwater in prediction and early warning system of tunnel geologic hazards. J. Shandong Univ. 39 (3), 115–121.

Liu, M., Liu, Z., Zhou, D., Lan, R., Wu, H., 2020. Recognition method of typical anomalies during karst tunnel construction using GPR attributes and Gaussian processes. Arab. J. Geosci. 13 (16), http://dx.doi.org/10.1007/s12517-020-05782-0.

Liu, B., Meng, W., Zhao, Z., Lin, T., Zhang, J., 2023b. Coupled thermal-hydro-mechanical modeling on characteristics of excavation damage zone around deep tunnels crossing a major fault. Tunn. Undergr. Space Technol. 134, 105008.

Liu, J.-Q., Sun, Y.-K., Li, C.-j., Yuan, H.-l., Chen, W.-Z., Liu, X.-Y., Zhou, X.-S., 2022. Field monitoring and numerical analysis of tunnel water inrush and the environmental changes. Tunn. Undergr. Space Technol. 122, http://dx.doi.org/10.1016/j.tust.2021.104360.

Liu, F., Zhang, W., Wang, G.-l., Wei, S.-c., Yue, C., Jiang, G.-z., Liao, Y.-z., 2023c. Geothermal anomalies in the xianshuihe area: Implications for tunnel construction along the Sichuan-Tibet Railway, China. J. Groundw. Sci. Eng. 11 (3), 237–248. http://dx.doi.org/10.26599/JGSE.2023.9280020.

Liu, W., Zhao, J., Nie, R., Liu, Y., Du, Y., 2018. A coupled thermal-hydraulic-mechanical nonlinear model for fault water inrush. Processes 6 (8), http://dx.doi.org/10.3390/pr6080120.

Luo, Y., Chen, J., Chen, Y., Diao, P., Qiao, X., 2018. Longitudinal deformation profile of a tunnel in weak rock mass by using the back analysis method. Tunn. Undergr. Space Technol. 71, 478–493. http://dx.doi.org/10.1016/j.tust.2017.10.003.

Nie, L., Song, Z., Zhou, W., Li, Z., Du, X., Ma, L., 2024. Full-decay induced polarization multi-parameter tomography method for tunnel water inrush ahead prospecting: A case study. Tunn. Undergr. Space Technol. 145, http://dx.doi.org/10.1016/j.tust.2024.105589.

Ogbu, A.D., Iwe, K.A., Ozowe, W., Ikevuje, A.H., 2024. Geostatistical concepts for regional pore pressure mapping and prediction. Glob. J. Eng. Technol. Adv. 20 (01), 105–117.

Olierook, H.K.H., Scalzo, R., Kohn, D., Chandra, R., Farahbakhsh, E., Clark, C., Reddy, S.M., Mueller, R.D., 2021. Bayesian geological and geophysical data fusion for the construction and uncertainty quantification of 3D geological models. Geosci. Front. 12 (1), 479–493. http://dx.doi.org/10.1016/j.gsf.2020.04.015.

Pan, D., Xu, Z., Lu, X., Zhou, L., Li, H., 2020. 3D scene and geological modeling using integrated multi-source spatial data: Methodology, challenges, and suggestions. Tunn. Undergr. Space Technol. 100, http://dx.doi.org/10.1016/j.tust.2020.103393.

Panzera, F., Alber, J., Imperatori, W., Bergamo, P., Fah, D., 2022. Reconstructing a 3D model from geophysical data for local amplification modelling: The study case of the upper Rhone valley, Switzerland. Soil Dyn. Earthq. Eng. 155, http://dx.doi.org/10.1016/j.soildyn.2022.107163.

Peng, X., Yu, P., Cheng, X., Chen, G., Zhang, Y., Zhang, H., Li, C., 2022. Dynamic simulation of the water inrush process in tunnel construction using a three-dimensional coupled discontinuous deformation analysis and smoothed particle hydrodynamics method. Tunn. Undergr. Space Technol. 127, http://dx.doi.org/10.1016/j.tust.2022.104612.

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A., 2018. CatBoost: unbiased boosting with categorical features. In: Bengio, S., Wallach, H., Larochelle, H., Grauman, K., CesaBianchi, N., Garnett, R. (Eds.), Advances in Neural Information Processing Systems 31. NIPS 2018, In: Advances in Neural Information Processing Systems, vol. 31, 32nd Conference on Neural Information Processing Systems (NIPS), Montreal, CANADA, DEC 02-08, 2018.

Rezaie, M., Moradzadeh, A., Kalate, A.N., Aghajani, H., Kahoo, A.R., Moazam, S., 2017. 3D modelling of trompsburg complex (in South Africa) using 3D focusing inversion of gravity data. J. Afr. Earth Sci. 130, 1–7. http://dx.doi.org/10.1016/j.jafrearsci.2017.03.002.

Ruben, v.d.G., Maarten, v.S., Dirk, T., Ben, V.C., 2022. The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression. J. Am. Med. Inform. Assoc. 29 (9), 1525–1534. http://dx.doi.org/10.1093/jamia/ocac093.

Rusu, C., Rusu, V., 2006. Radial basis functions versus geostatistics in spatial interpolations. In: Bramer, M. (Ed.), Artificial Intelligence in Theory and Practice. Springer US, Boston, MA, pp. 119–128.

Shen, Y., Ling, J., Li, X., Li, H., Feng, S., Zhu, H., 2022. Holistic digital-twin-based framework to improve tunnel lighting environment: From methodology to application. Build. Environ. 224, 109562.

Shi, S., Bu, L., Li, S., Xiong, Z., Xie, X., Li, L., Zhou, Z., Xu, Z., Ma, D., 2017. Application of comprehensive prediction method of water inrush hazards induced by unfavourable geological body in high risk karst tunnel: a case study. Geomat. Nat. Hazards Risk 8 (2), 1407–1423. http://dx.doi.org/10.1080/19475705.2017.1337656.

Shi, Y., Xu, F., Song, X., Wang, G., Zuo, Y., Li, X., Ji, J., 2023. Rock damage evolution in the production process of the enhanced geothermal systems considering thermal-hydrological-mechanical and damage (THM-D). Energy 285, http://dx.doi.org/10.1016/j.energy.2023.129421.

Standard, C., 2004. Code for design of road tunnel.

Su, G., Peng, L., Hu, L., 2017. A Gaussian process-based dynamic surrogate model for complex engineering structural reliability analysis. Struct. Saf. 68, 97–109. http://dx.doi.org/10.1016/j.strusafe.2017.06.003, URL: https://www.sciencedirect.com/science/article/pii/S016747301730214X.

Sun, J., Wang, F., Wang, X., Wu, X., 2021. A quantitative evaluation method based on back analysis and the double-strength reduction optimization method for tunnel stability. Adv. Civ. Eng. 2021, http://dx.doi.org/10.1155/2021/8899685.

Sun, H., Zhong, D., Wu, Z., Wang, L., 2024. Multi-labeled regularized marching tetrahedra method for implicit geological modeling. Math. Geosci. 56 (2), 219–248. http://dx.doi.org/10.1007/s11004-023-10075-9.

Tao, F., Xiao, B., Qi, Q., Cheng, J., Ji, P., 2022. Digital twin modeling. J. Manuf. Syst. 64, 372–389.

The Compilation Committee of Engineering Geology Handbook, 2007. Engineering Geology Handbook 4th Edition. China Architecture and Building Press, Beijing.

Wang, J., Li, P., Li, X., Zhu, H., 2021. Complex 3D geological modeling based on digital twin. IOP Conf. Ser.: Earth Environ. Sci. 861 (7), 072046. http://dx.doi.org/10.1088/1755-1315/861/7/072046.

Wang, C., Liu, Z., Zhang, F., Guo, Q., Dong, Z., Bai, P., 2024a. Heat hazards in high-temperature tunnels: Influencing factors, disaster forms, the geogenetic model and a case study of a tunnel in southwest China. Sustainability 16 (3), http://dx.doi.org/10.3390/su16031044.

Wang, F., Zhang, D., Huang, H., Huang, Q., 2023. A phase-field-based multi-physics coupling numerical method and its application in soil-water inrush accident of shield tunnel. Tunn. Undergr. Space Technol. 140, http://dx.doi.org/10.1016/j.tust.2023.105233.

Wang, C., Zhang, D., Tan, D., Ye, J., Wang, X., Ren, F., Li, Q., 2024b. Investigation on seepage evolution property of surrounding rock under the tunneling and water surge. Tunn. Undergr. Space Technol. 153, 105970. http://dx.doi.org/10.1016/j.tust.2024.105970.

Wang, J., Zhao, H., Bi, L., Wang, L., 2018. Implicit 3D modeling of ore body from geological boreholes data using Hermite radial basis functions. Minerals 8 (10), http://dx.doi.org/10.3390/min8100443.

Wu, A., Zhao, W., Zhou, H., Liu, F., Gong, G., Wang, B., 2014. Standard for engineering classification of rock mass.

Wu, H., Zhu, Q., Guo, Y., Zheng, W., Zhang, L., Wang, Q., Zhou, R., Ding, Y., Wang, W., Pirasteh, S., et al., 2022. Multi-level voxel representations for digital twin models of tunnel geological environment. Int. J. Appl. Earth Obs. Geoinf. 112, 102887.

Xiaojun, L., Li, T., Jiaxin, L., et al., 2023. Digital-twin-enabled JIT design of rock tunnel: Methodology and application. Tunn. Undergr. Space Technol. Incorp. Trenchless Technol. Res. 140.

Xie, Q., Cao, Z., Sun, W., Fumagalli, A., Fu, X., Wu, Z., Wu, K., 2023. Numerical simulation of the fluid-solid coupling mechanism of water and mud inrush in a water-rich fault tunnel. Tunn. Undergr. Space Technol. 131, http://dx.doi.org/10.1016/j.tust.2022.104796.

Xin-rong, L., Yong-quan, L., Zhong-ping, Y., Yi-liang, T., 2015. Synthetic advanced geological prediction technology for tunnels based on GPR. Chin. J. Geotech. Eng. 37 (zk2), 51–56.

Xiong, Z., Guo, J., Xia, Y., Lu, H., Wang, M., Shi, S., 2018. A 3D multi-scale geology modeling method for tunnel engineering risk assessment. Tunn. Undergr. Space Technol. 73, 71–81. http://dx.doi.org/10.1016/j.tust.2017.12.003.

Yu, G., Wang, Y., Mao, Z., Hu, M., Sugumaran, V., Wang, Y.K., 2021. A digital twin-based decision analysis framework for operation and maintenance of tunnels. Tunn. Undergr. Space Technol. 116, 104125.

Zhai, S., Zhan, J., Ba, Y., Chen, J., Li, Y., Li, Z., 2019. PFC model parameter calibration using uniform experimental design and a deep learning network. IOP Conf. Ser.: Earth Environ. Sci. 304 (3), 032062. http://dx.doi.org/10.1088/1755-1315/304/3/032062.

Zhang, Y., Su, G., Liu, B., Li, T., 2020. A novel displacement back analysis method considering the displacement loss for underground rock mass engineering. Tunn. Undergr. Space Technol. 95, 103141. http://dx.doi.org/10.1016/j.tust.2019.103141.

Zhao, C., Lei, Q., Zhang, Z., Loew, S., 2023. Causal mechanism of gotthard base tunnel-induced ground deformation: Insights from 3D fully-coupled hydro-mechanical simulation and comparison to field measurements. Int. J. Rock Mech. Min. Sci. 170, http://dx.doi.org/10.1016/j.ijrmms.2023.105515.

Zhao, J., Liu, W., Zheng, Q., Sun, X., 2020. Study on a coupled thermal-hydraulic-mechanical model for water inrush from collapse column. Geotech. Geol. Eng. 38 (5), 4761–4766. http://dx.doi.org/10.1007/s10706-020-01325-2.

Zheng, F., Jiang, A., Guo, X., Min, Q., Yin, Q., 2022. Back analysis of surrounding rock parameters of large-span arch cover station based on GP-DE algorithm. Appl. Sci. 12 (24), http://dx.doi.org/10.3390/app122412590.

Zhong, D.-Y., Wang, L.-G., Wang, J.-M., 2021. Combination constraints of multiple fields for implicit modeling of ore bodies. Appl. Sci. 11 (3), http://dx.doi.org/10.3390/app11031321.

Zhou, Z., Li, L., Shi, S., Liu, C., Gao, C., Tu, W., Wang, M., 2020. Study on tunnel water inrush mechanism and simulation of seepage failure process. Rock Soil Mech. 41 (11), 3621–3631. http://dx.doi.org/10.16285/j.rsm.2020.0131.

Zhu, J., Zou, H., Rosset, S., Hastie, T., 2009. Multi-class AdaBoost. Statist. Interface 2 (3), 349–360.